

On-Policy Distillation for LLM Post-Training

Principles, Advantages, and Insights

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Background: OPD Goes Mainstream

- ▶ Three stages of LLM training: Pre-training → Mid-training → **Post-training**
- ▶ OPD has moved from a research idea to a **standard post-training ingredient**:
 - Qwen3: explicit OPD stage — AIME'24 **67.6% → 74.4%**, GPU-hours **17,920 → 1,800**
 - DeepSeek-V4: distillation + RL hybrid post-training recipes
 - GLM-5, Xiaomi MiMo (MiMo-V2-Flash): on-policy (self-)distillation for multi-capability integration
- ▶ **OPD is becoming the third pillar** of post-training, next to SFT and RLHF/RLVR



2025-05-15

Qwen3 Technical Report

Qwen Team

4.5 Strong-to-Weak Distillation

The Strong-to-Weak Distillation pipeline is specifically designed to optimize lightweight models, encompassing 5 dense models (Qwen3-0.6B, 1.7B, 4B, 8B, and 14B) and one MoE model (Qwen3-30B-A3B). This approach enhances model performance while effectively imparting robust mode-switching capabilities. The distillation process is divided into two primary phases:

- (1) **Off-policy Distillation**: At this initial phase, we combine the outputs of teacher models generated with both /think and /no.think modes for response distillation. This helps lightweight student models develop basic reasoning skills and the ability to switch between different modes of thinking, laying a solid foundation for the next on-policy training phase.
- (2) **On-policy Distillation**: In this phase, the student model generates on-policy sequences for fine-tuning. Specifically, prompts are sampled, and the student model produces responses in either /think or /no.think mode. The student model is then fine-tuned by aligning its logits with those of a teacher model (Qwen3-32B or Qwen3-235B-A22B) to minimize the KL divergence.



DeepSeek-V4:

Towards Highly Efficient Million-Token Context Intelligence

DeepSeek-AI
research@deepseek.com

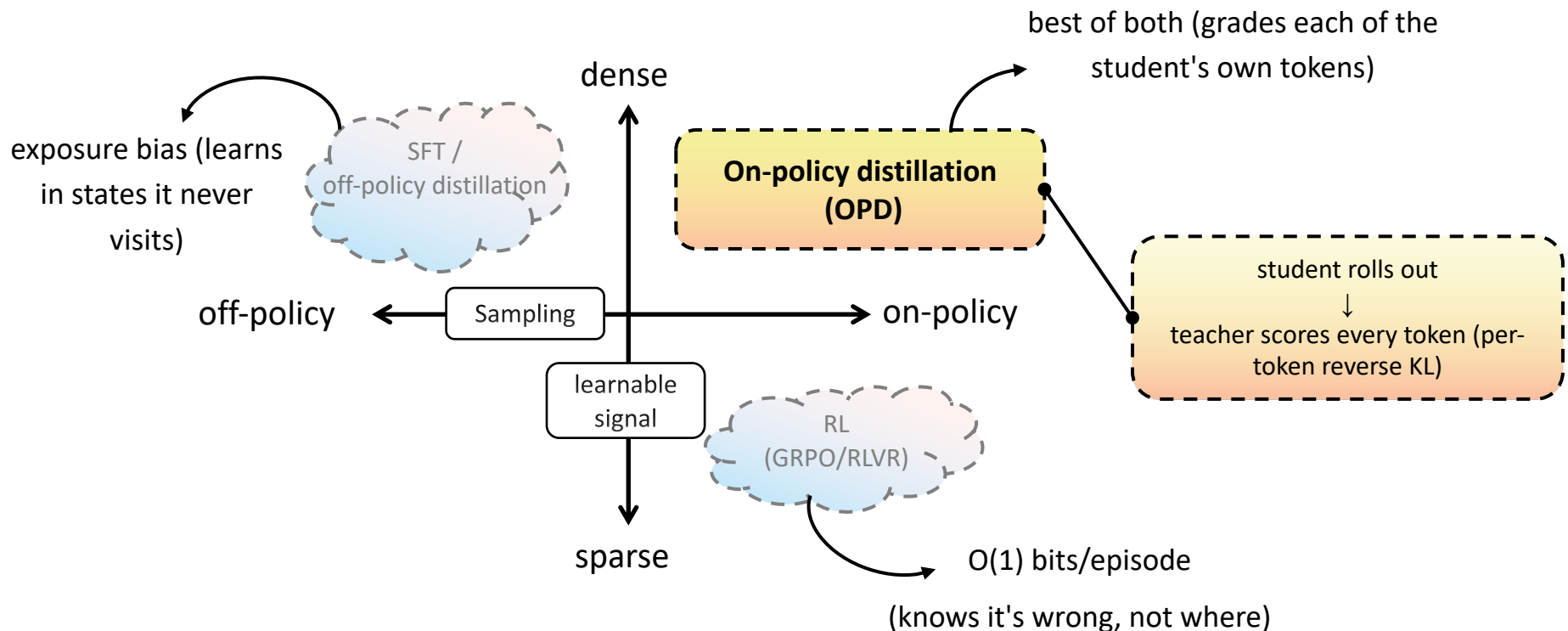
The post-training pipeline of DeepSeek-V4 series features a two-stage paradigm: the independent cultivation of domain-specific experts, followed by unified model consolidation via on-policy distillation (Gu et al., 2024; Lu and Lab, 2025). Initially, for each target domain — such

GLM-5: from Vibe Coding to Agentic Engineering

pipeline—starting with Reasoning RL, followed by Agentic RL, and finishing with General RL. Crucially, we utilized On-Policy Cross-Stage Distillation throughout this process to prevent catastrophic forgetting, ensuring the model retains its sharp reasoning edge while becoming a robust generalist. In summary, the leap in GLM-5's performance is driven by the following technical contributions:

Introducing OPD: The Best of Both Worlds

- ▶ **Recap:** Every post-training method is a choice on two axes:
 - ▶ **Sampling** — off-policy (imitate external data) vs on-policy (learn from own rollouts)
 - ▶ **Learnable signal** — sparse (sequence) vs dense (token-wise)



Introducing OPD: The Best of Both Worlds

► Chess analogy:

- **On-policy RL** = playing with no coach -> feedback tied to your own play, but only 1 bit per game
- **Off-policy distillation (SFT)** = watching a grandmaster -> very strong moves, but often in states a novice rarely reaches
- **OPD** = playing with no coach + a grandmaster rates each move -> feedback for each move of your own



How OPD Works

- ▶ **Pipeline:**

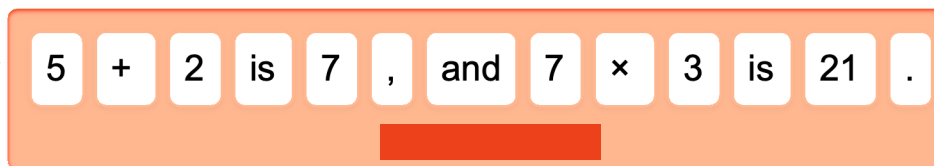
student samples a trajectory → teacher grades it **token-by-token** (reverse KL) → per-token KL becomes a negative reward → RL-style policy-gradient update

- ▶ Teacher only needs **one forward pass**: no generation needed
- ▶ Training can use short / partial rollouts; no separate reward model needed

prompt

What is $5 + (2 \times 3)$?

student trajectory



student trajectory



teacher's conditional per-token probability

Early Evidence: Numbers from Qwen3

Stage	AIME'24	GPQA-Diamond	GPU-hours
Off-policy distillation (SFT)	55.0%	55.6%	—
+ Reinforcement Learning	67.6%	61.3%	17,920
+ On-Policy Distillation	74.4%	63.3%	1,800 (~10× less)

► Source: Qwen3 Technical Report, Table 21

Part 1 Recap

- ▶ OPD = on-policy sampling + dense per-token teacher supervision
= RL's **relevance** + SFT's **information density**
- ▶ Already a standard post-training step at Qwen3, DeepSeek, GLM, MiMo
- ▶ Next part: **math behind this, and why does it deliver order-of-magnitude efficiency gains?**

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Notation Setup

- ▶ Prompt x
- ▶ Student policy: π_θ
- ▶ Teacher policy: π_{teacher} (frozen)
- ▶ On-policy rollout: $\hat{y} = (\hat{y}_1, \dots, \hat{y}_T) \sim \pi_\theta(\cdot, x)$
- ▶ Prefix: $c_t = (x, \hat{y})$

- ▶ Reverse KL between student and teacher, evaluated on the student's own next token:

$$\text{KL}(\pi_\theta \parallel \pi_{\text{teacher}}) = \mathbb{E}_{\hat{y} \sim \pi_\theta} \left[\log \pi_\theta(\hat{y}_{t+1} \mid c_t) - \log \pi_{\text{teacher}}(\hat{y}_{t+1} \mid c_t) \right]$$

Three Objectives Side by Side

- ▶ **SFT** (off-policy, MLE) -> no notion of the student's own distribution:

$$\mathcal{L}_{\text{SFT}}(\theta) = - \mathbb{E}_{(x, y^*) \sim \mathcal{D}} [\log \pi_{\theta}(y^* \mid x)]$$

- ▶ **GRPO** (on-policy, sparse) -> A_i is one number per trajectory:

$$J_{\text{GRPO}}(\theta) = \mathbb{E} \left[\frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min(\rho_{i,t} A_i, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) A_i) \right] \\ - \beta D_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}})$$

$$A_i = \frac{r_i - \text{mean}(\{r_1, \dots, r_G\})}{\text{std}(\{r_1, \dots, r_G\})}$$

- ▶ **OPD** (on-policy, dense) -> an independent signal at every t:

$$J_{\text{OPD}}(\theta) = \mathbb{E}_x [\text{KL}(\pi_{\theta}(\cdot \mid x) \parallel \pi_{\text{teacher}}(\cdot \mid x))]$$

Deriving the OPD Gradient

- Differentiate J_{OPD} — product rule on the sampling distribution π_θ :

$$\begin{aligned}\nabla_\theta \mathbb{E}_{\hat{y} \sim \pi_\theta(\cdot | x)} [f_\theta(\hat{y})] &= \mathbb{E}_{\hat{y} \sim \pi_\theta} [f_\theta(\hat{y}) \nabla_\theta \log \pi_\theta(\hat{y} | x)] + \mathbb{E}_{\hat{y} \sim \pi_\theta} [\nabla_\theta f_\theta(\hat{y})] \\ &\text{with } f_\theta(\hat{y}) = \log \pi_\theta(\hat{y} | x) - \log \pi_{\text{teacher}}(\hat{y} | x) \\ \implies \nabla_\theta f_\theta(\hat{y}) &= \nabla_\theta \log \pi_\theta(\hat{y} | x) \quad (\pi_{\text{teacher}} \text{ is frozen})\end{aligned}$$

- The score function has zero mean $\Rightarrow$ the θ -dependence inside the reward drops out:

$$\begin{aligned}\mathbb{E}_{\hat{y} \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(\hat{y} | x)] &= \nabla_\theta \int \pi_\theta(\hat{y} | x) d\hat{y} = \nabla_\theta 1 = 0 \\ \implies \nabla_\theta J_{\text{OPD}} &= \mathbb{E}_{x, \hat{y} \sim \pi_\theta} \left[\left(\log \pi_\theta(\hat{y} | x) - \log \pi_{\text{teacher}}(\hat{y} | x) \right) \right. \\ &\quad \left. \times \nabla_\theta \log \pi_\theta(\hat{y} | x) \right]\end{aligned}$$

- Autoregressive decomposition, keep only $t' = t$ (TML's “discount = 0”):

$$\begin{aligned}\log \pi_\theta(\hat{y} | x) &= \sum_{t=1}^T \log \pi_\theta(\hat{y}_t | c_t) \\ \nabla_\theta J_{\text{OPD}} &\approx \mathbb{E}_{x, \hat{y} \sim \pi_\theta} \left[\sum_{t=1}^T \left(\log \pi_\theta(\hat{y}_t | c_t) - \log \pi_{\text{teacher}}(\hat{y}_t | c_t) \right) \right. \\ &\quad \left. \times \nabla_\theta \log \pi_\theta(\hat{y}_t | c_t) \right]\end{aligned}$$

Comparison: OPD vs. RL(GRPO) vs. SFT

Dimension	SFT (off-policy)	GRPO / RLVR	On-Policy Distillation
Objective	$-\log \pi_{\theta}(y^* x)$	Clipped PG, adv. A_i	$D_{KL}(\pi_{\theta} \pi_{teacher})$, per-token
Sampling	off-policy	on-policy	on-policy
Signal density	dense (hard labels)	sparse (sequence)	dense (per-token)
Teacher needed	implicit (data gen)	none	yes, one fwd pass
Typical failure	exposure bias	sparse gradient	teacher capability ceiling
Training cost	low	high (long rollouts)	medium (teacher fwd pass)
Best for	cold start, expert data	verifiable-reward tasks	compress / recover capability

Forward KL vs. Reverse KL

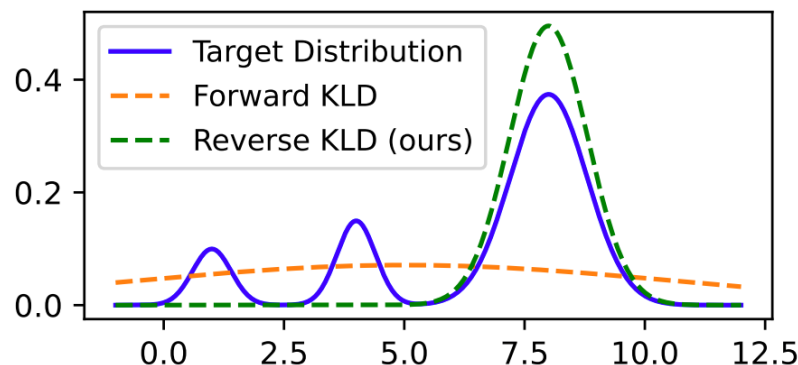
► KL divergence is **not symmetric**: $D_{\text{KL}}(p||q) \neq D_{\text{KL}}(q||p)$ in general — which one you pick changes the answer completely

Forward KL — average log-ratio under samples from p : $D_{\text{KL}}(p || q) = \mathbb{E}_{x \sim p} [\log p(x) - \log q(x)]$

Reverse KL — average log-ratio under samples from q : $D_{\text{KL}}(q || p) = \mathbb{E}_{x \sim q} [\log q(x) - \log p(x)]$

► **Forward KL($p||q$)** — if $q(x) \rightarrow 0$ where $p(x) > 0$, then $\log(p/q) \rightarrow \infty$: infinite penalty. So minimizing it forces **q to cover every place p has mass** — “mass-covering” for q .

► **Reverse KL($q||p$)** — if $q(x) > 0$ where $p(x) \rightarrow 0$, then $\log(q/p) \rightarrow \infty$: infinite penalty. So minimizing it forces **q to vanish wherever p vanishes**, letting q ignore some modes of p and commit to others — “mode-seeking” for q .



Why OPD Chooses Reverse KL

- ▶ SFT $\Leftrightarrow$ forward $KL(\pi_{\text{teacher}}||\pi_{\theta})$ — off-policy, mass-covering. OPD $\Leftrightarrow$ reverse $KL(\pi_{\theta}||\pi_{\text{teacher}})$ — on-policy, mode-seeking.
- ▶ Mode-seeking $\Rightarrow$ the student commits to **one** coherent, teacher-approved path, not an incoherent average over several
- ▶ Reverse KL is also “**unhackable**”: graded on what the student actually did — no black-box reward model to game
- ▶ **Evidence** — Gu et al., *MiniLLM* (2023, arXiv:2306.08543), GPT-4 feedback score (0–100), LLaMA-7B student vs. LLaMA-13B teacher (Table 1):

Method (7B student)	DollyEval	SelfInst	VicunaEval
Teacher (13B)	79.0	75.5	65.1
SFT (no distillation)	73.0	69.2	61.6
KD — forward KL	73.7	70.5	62.7
MiniLLM — reverse KL	76.4	73.1	64.1

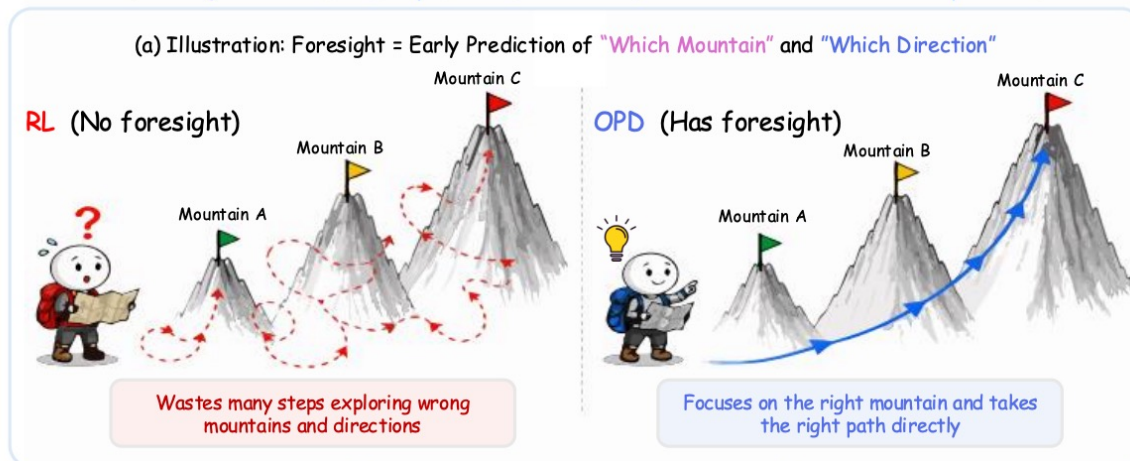
Why Efficient? View 1: Information Theory & Search-vs-Shortcut

- ▶ RL teaches **$O(1)$ bit** / episode; distillation teaches **$O(N)$ bits** / episode (N = tokens)
- ▶ RL spends compute mostly on **search** (rollout + credit assignment), not the update itself
- ▶ Once RL finds a good strategy, distillation is a **shortcut** to learn it
- ▶ Analogy: teaching a scientific result in words is cheap; teaching a physical skill (muscle memory) is expensive
- ▶ Evidence: OPD replicates an RL-trained policy in **7–10× fewer gradient steps**

Why Efficient? View 2: Parameter Dynamics — “Foresight”

- ▶ Question: given the same init, how do ΔW_{OPD} and ΔW_{RL} differ?
- ▶ Answer: OPD locks onto a stable, final-solution-aligned update trajectory **early** in training — “Foresight”

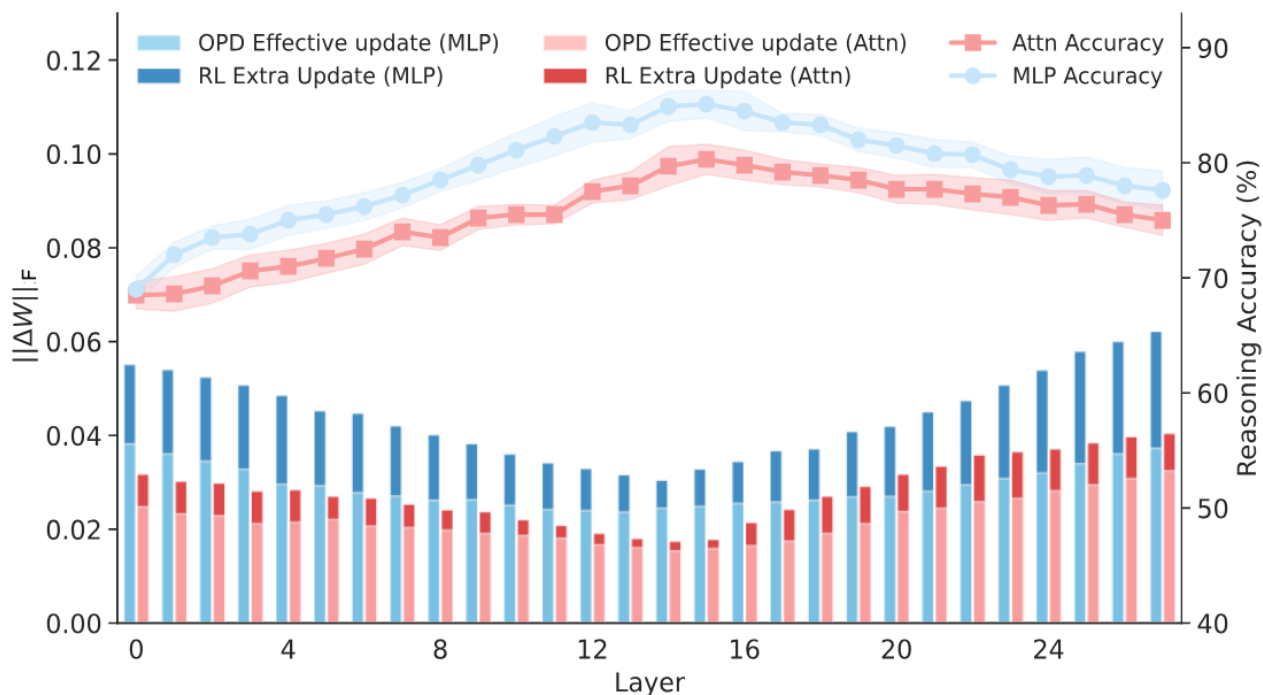
Part 1: Foresight in OPD - Early Determination of Module-Allocation and Update-Directions



- ▶ **Property 1 — Functional Redundancy Avoidance:** updates concentrate in reasoning-critical mid-layer MLPs
- ▶ **Property 2 — Early Low-Rank Lock-in:** update energy concentrates in a few dominant directions, aligned with the final subspace from very early on

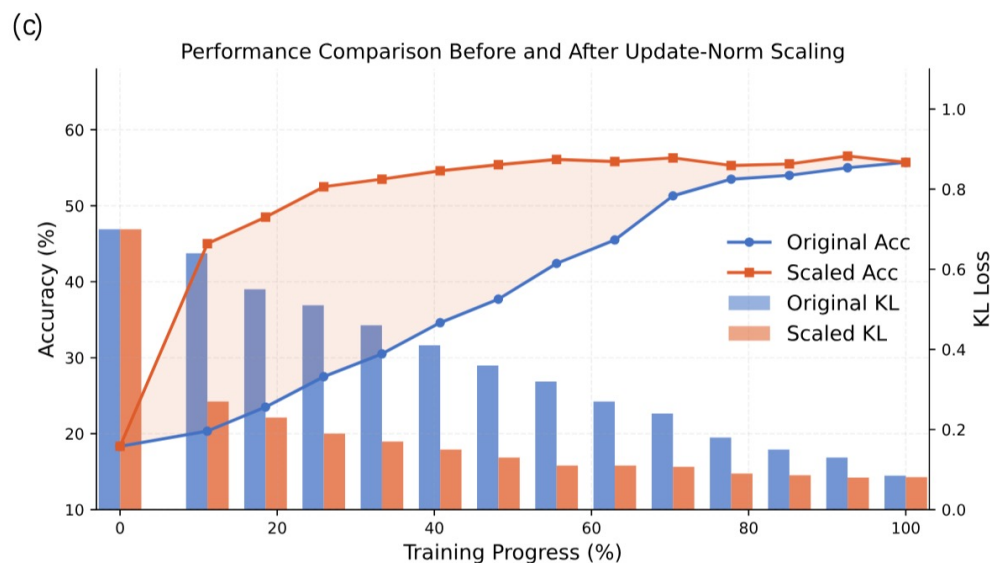
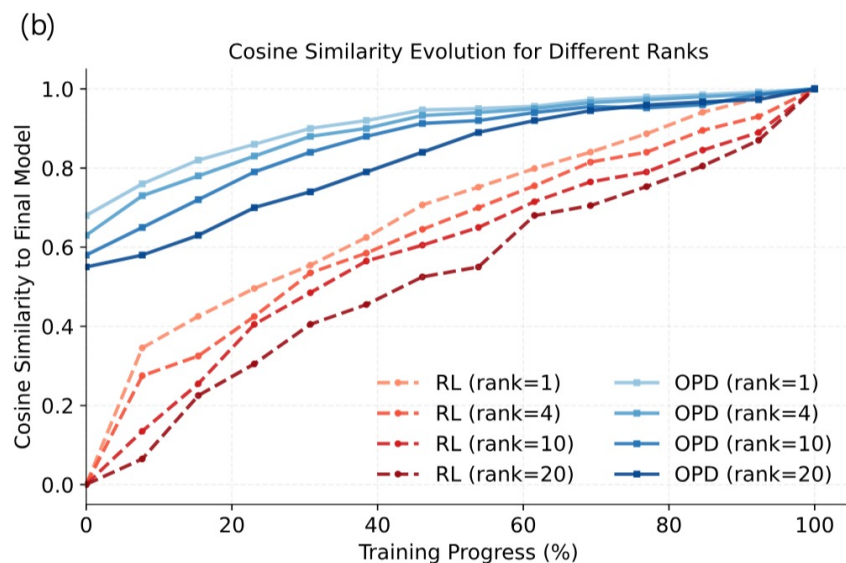
Evidence for Property 1: Functional Redundancy Avoidance

- ▶ Claim: OPD concentrates updates on reasoning-critical modules; RL wastes update norm on low-utility ones — both use the *same* functional pathway, allocated differently
- ▶ **Sliding-window intervention** (Qwen3-8B-Base, MATH500): both RL and OPD show the *same* inverted-U sensitivity by layer depth (middle-layer MLPs matter most) — but **RL's update norm in the low-sensitivity top/bottom layers is far larger than OPD's**, for no matching accuracy gain



Evidence for Property 2: Early Low-Rank Lock-in

- ▶ Claim: OPD's update energy concentrates in a few dominant directions, and those directions already match the **final** model early in training
- ▶ **Fig. 5b–c (below)**: OPD's dominant-subspace alignment with the final model is already high in the **first 0–30%** of training; rescaling a **10%-progress** checkpoint's update to the final norm (same direction, more magnitude) recovers **~80%** of final performance



- ▶ Takeaway: for OPD, most of training after the first ~10–30% is really just **turning up the volume** on a direction that was already right

Part 2 Recap

- ▶ OPD replaces SFT's hard labels and RL's sparse sequence-level reward with **per-token reverse KL**
- ▶ Efficiency has two complementary explanations: information-theoretic shortcut (macro) + parameter-dynamics Foresight (micro)
- ▶ Next: **two variants of OPD**

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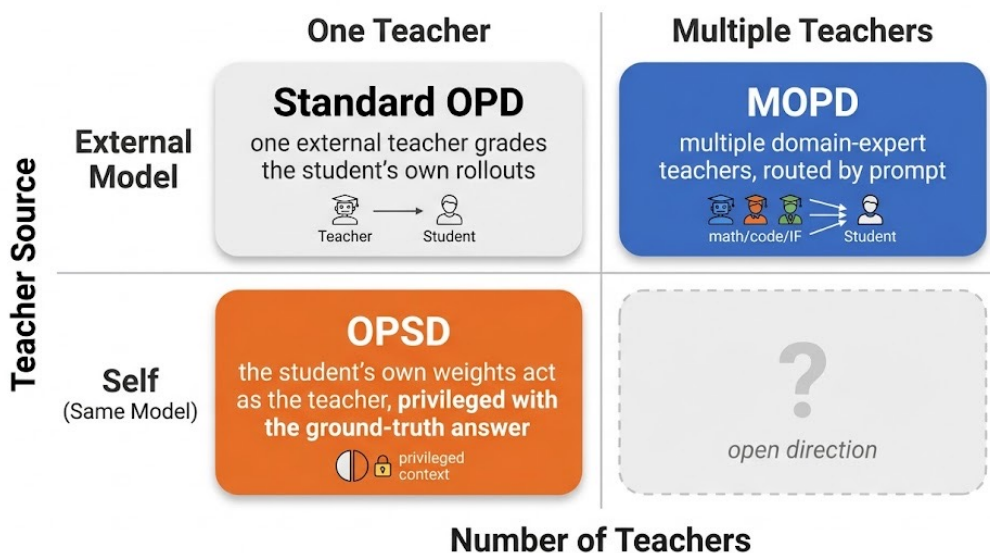
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Beyond a Single Teacher: Two Frontiers of OPD

- ▶ Standard OPD assumes **one** external teacher for **one** capability. Two natural extensions:
- ▶ **More teachers, one student** → Multi-Teacher On-Policy Distillation (MOPD): integrate several domain experts (math / code / SWE / IF) into one model
- ▶ **No external teacher at all** → On-Policy Self-Distillation (OPSD): the model becomes its own teacher, using the ground-truth answer as privileged context



MOPD Motivation: Why Multi-Teacher OPD?

- ▶ Math, code, SWE, instruction-following each need very different RL recipes → strong single-domain experts, but how to merge into one model?

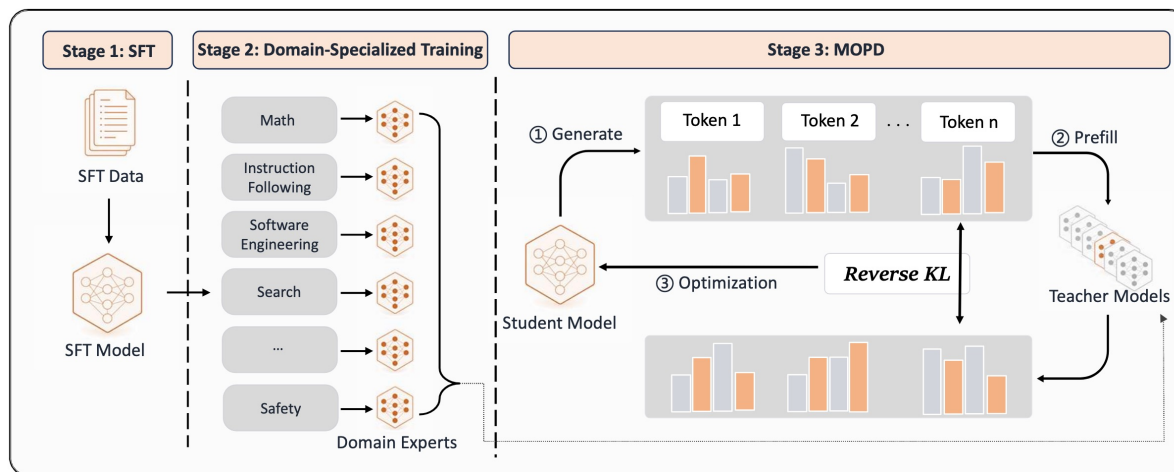
Approach	Core Problem
MixRL (joint mixed RL)	cross-domain interference (“see-saw”), not parallelizable
SeqRL (sequential RL)	later domains erode earlier capabilities
RFT (offline SFT on teacher rollouts)	exposure bias
PM (parameter merging)	unstable in weight space

- ▶ **MOPD**: the only approach that is dense + on-policy + parallelizable, **simultaneously**

MOPD Method: Three-Stage Pipeline & Routing

- ▶ **Stage 1 — General SFT:** shared initialization checkpoint
- ▶ **Stage 2 — Domain-specialized RL:** each domain trains its own frozen expert from the same Stage-1 checkpoint, in parallel (**same-origin teachers**)
- ▶ **Stage 3 — MOPD:** student rolls out on its own; each prompt is **routed by domain** to its matching expert; minimize per-token reverse KL

$$\mathcal{L}_{\text{rev-KL}} = \mathbb{E}_{x, y \sim \pi_{\theta}} \left[\frac{1}{|y|} \sum_t \sum_v \pi_{\theta}(v) \log \frac{\pi_{\theta}(v)}{\pi_{\phi_d}(v)} \right]$$



MOPD Method (cont'd): Top-K Distillation & Co-evolving Teachers

$$\mathcal{L}_{\text{MOPD}}^{\text{TopK}}(\theta) = \mathbb{E}_{x,y} \left[\frac{1}{|y|} \sum_t \sum_{v \in \mathcal{T}_t^d} \left(\pi_{\theta}(v) \log \frac{\pi_{\theta}(v)}{\pi_{\phi_d}(v)} - \pi_{\theta}(v) + \pi_{\phi_d}(v) \right) \right]$$
$$\mathcal{T}_t^d = \text{TopK}_k(\pi_{\phi_d}(\cdot \mid x, y_{<t}))$$

- ▶ Cheaper alternative to full-vocabulary distillation (default $k = 64$)
- ▶ **Multi-round co-evolution:** retrain domain RL teachers from the new (round-1) student, then run round-2 MOPD
- ▶ On Qwen3: one such round (Math + IF) improved the normalized score **0.937** $\rightarrow$ **0.986**; the round-2 RL teacher itself reached **1.030**

MOPD Results: Qwen3-30B-A3B (Math / IF / SWE)

Method	AIME25	AIME26	IFBench	IFEval	SWE-bench V.	Norm. Score
MixRL	52.71	63.75	75.00	94.58	48.80	0.882
SeqRL	48.54	61.88	77.11	95.80	47.80	0.775
RFT	51.56	63.44	80.95	93.35	45.80	0.824
PM	49.38	63.96	78.23	95.81	48.80	0.857
MOPD (ours)	51.46	65.31	77.89	93.84	50.40	0.937

- ▶ Normalized score: 0 = SFT-student level, 1 = RL-expert level
- ▶ MOPD closes **91–95%** of the gap across all three domains — most balanced (spread only 0.044 vs. SeqRL's 0.41)
- ▶ **Deployed in production:** MiMo-V2-Flash (~309B) — matches/exceeds RL teacher on 5 of 7 benchmarks
 - AIME25 94.1, HMMT25 84.4, LCB 83.2, τ^2 -Bench 80.3, τ^2 -Telecom 95.3

OPSD Motivation: Why On-Policy Self-Distillation?

- ▶ Standard OPD's “sweet burden”: needs a **separate, usually larger** teacher to serve/host
- ▶ Doesn't explicitly use the ground-truth solutions already sitting in reasoning datasets
- ▶ Key insight (evaluation $\geq$ generation): **rationalizing a known answer is easier than generating it from scratch**
- ▶ OPSD's idea: **one model, two roles** via context —
 - Student $p_S(\cdot|x)$: sees **only the problem**
 - Teacher $p_T(\cdot|x,y^*)$: sees the problem + **the ground-truth solution** (privileged info)

OPSD Method: Core Formulation

- Same parameters θ , different conditioning:

$$p_T(\cdot \mid x, y^*) \triangleq p_\theta(\cdot \mid x, y^*). \quad p_S(\cdot \mid x) \triangleq p_\theta(\cdot \mid x).$$

$$\mathcal{L}(\theta) = \mathbb{E}_{(x, y^*)} \mathbb{E}_{\hat{y} \sim p_S(\cdot \mid x)} \left[\frac{1}{|\hat{y}|} \sum_{n=1}^{|\hat{y}|} D(p_T(\cdot \mid x, y^*, \hat{y}_{<n}) \parallel p_S(\cdot \mid x, \hat{y}_{<n})) \right]$$

Student Prompt

Problem: Find the derivative of $f(x) = 3x^2 + 2x - 5$ at $x = 2$

Answer:

Teacher Prompt

Problem: Find the derivative of $f(x) = 3x^2 + 2x - 5$ at $x = 2$

Here is a reference solution:

First find $f'(x) = 6x + 2$, then evaluate at $x = 2$: $f'(2) = 6(2) + 2 = 14$

After understanding the reference solution, please try to solve this problem using your own approach below:

Answer:

- Teacher is fixed to the initial model rather than the live one -> stabilizes training.

OPSD Results: Qwen3-Instruct on Math Reasoning

Model	Base	+SFT	+GRPO	+OPSD
1.7B	37.1	35.8 ↓	37.7	43.4 (+6.3)
4B	61.2	58.6 ↓	62.7	63.6
8B	61.8	59.8 ↓	64.0	64.8

- ▶ **SFT hurts:** overly concise reference solutions shorten test-time reasoning
- ▶ **GRPO stalls:** >50% of batches have zero within-group reward variance within 100 steps
- ▶ **Efficiency:** OPSD = 1 rollout × 1,024 tokens × 100 steps; GRPO = 8 rollouts × 16K tokens × 500 steps

Full Landscape: SFT / GRPO / OPD / OPSD / MOPD

Dimension	SFT	GRPO	OPD	OPSD	MOPD
Sampling	off-policy	on-policy	on-policy	on-policy	on-policy
Signal density	dense (labels)	sparse (seq.)	dense (token)	dense (token)	dense (token)
Teacher	implicit	none	external	self	multi-expert (same-origin)
Uses GT answer	as target	as verifier	no	yes (privileged)	no
Solves	cold start	verifiable RL	compress/recover	remove ext. teacher	multi-domain integration

Summary

- ▶ **OPD** = on-policy sampling + dense per-token supervision — the sweet spot between RL and SFT
- ▶ **Efficiency**: information-theoretic shortcut (macro) + parameter-dynamics Foresight (micro)
- ▶ **Two natural extensions**: **MOPD** (multi-expert integration) and **OPSD** (no external teacher)
- ▶ **Open directions**:
 - OPD + reward-hacking robustness at scale;
 - a unified theory linking distillation objectives, evolving on-policy distributions, and parameter spectra;
 - extending to agentic / multi-turn / multi-modal settings;
 - Foresight-based adaptive acceleration

Thank You

Questions & Discussion

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